Manas Mahale

manas.m.mahale@gmail.com · github.com/Manas02 · manasmahale.in

Experience

July 2023–Present

Group Member

The Bender Group, University of Cambridge, UK & BBU, RO

Developed Ligand-Target Prediction models for phenotypic screening. Researching Chemically meaningful validation using Biased data splitting and Foundational Models for Single Cell Perturbation data.

Jan 2026–June 2026

Group Member

Photochemistry Group, Universidade Nova de Lisboa, PT

Characterising novel supramolecular synthetic biomimetic systems with photochemical properties.

Aug 2022–June 2023

AI & ML Intern

Pangea Botanica, Berlin, DE

Created Ligand-Target Prediction suite, and deployed computational metabolomics modules on AWS. Optimized MS2DeepScore for Natural Product similarity. Developed computational Pharmacokinetic models.

Feb 2022–June 2023

Affiliate Undergraduate Researcher

Center for Molecular Informatics, University of Cambridge, UK

Investigated methods to predict structural similarity given MS/MS data using Positional Embeddings and Attention. Explored Masked Language Models (BERT) for constrained small molecule library generation.

June 2022–July 2022

Computational Chemistry Consultant

ChemBio Discovery, Boston, USA

Automated Schrödinger workflows for the computational validation of Covalent Docking Pipelines, reducing processing time by 75%. Streamlined protein alignment and homology modeling processes, enhancing accuracy and efficiency.

Aug 2021–Aug 2022

Undergraduate Researcher

The Coutinho Lab, Mumbai, IN

Developed Chemical Language Models for Fragment Based Drug Design and Analogue Split as a chemically biased parametric data splitting method for model validation against IID activity cliff data.

Open Source Software

`pip install dilipred`

DILIPredictor

github.com/Manas02/dili-pip

Developed a package for in-silico prediction of drug-induced liver injury (DILI). Achieved an AUC-PR of 0.79 enabling improved detection of hepatotoxic compounds in humans.

`pip install analoguesplit`

Analogue Split

github.com/Manas02/analogue-split

Validates QSAR model robustness in handling activity cliffs. Introduced $\gamma-$plot visualization for model diagnostics across test set cases.

`pip install fbdd`

FragmentBERT

github.com/Manas02/fbdd

BERT-based Masked Language Model trained on tokens of drug-like compound's substructures. Retrospectively validated fragment linking, merging and growth modalities using virtual screening.

`pip install molecularnetwork`

MolecularNetwork

github.com/Manas02/molecularnetwork

A tool to create molecular similarity networks using RDKit and NetworkX, with customizable descriptors for graph-based SAR exploration.

`pip install pksmart`

PKSmart

github.com/Manas02/pksmart-web

Developed a package and webapp that models human PK parameters (VD_{ss}, Cl, f_u) using structural fingerprints, physicochemical descriptors and predicted animal data. Achieved prediction accuracies comparable to industry-standard models for key PK parameters.

Publications

2026 Bender, A. et al. *Artificial intelligence in drug discovery — what it is, where we stand and the path forward*. Nat Rev Drug Discov. https://doi.org/10.1038/s41573-026-01496-2.

2026 Mahale, M. et al. *The Medicinal Chemist's Map to Deep Learning: Concepts, Applications, and Case Studies*. In Reference Module in Chemistry, Molecular Sciences and Chemical Engineering. Elsevier. https://doi.org/10.1016/B978-0-443-29808-0.00066-2.

2026 Fernandes, C.; Mahale, M.; Thakur, S.; Patel, P.; Kaul, R. *Immunology Biomarkers in Cancer*. In Role of Biomarkers in Cancer Diagnosis and Therapeutics; Elsevier; pp 231–256. https://doi.org/10.1016/B978-0-443-44041-0.00020-9.

2025 Seal, S. et al. *PKSmart: an open-source computational model to predict intravenous pharmacokinetics of small molecules*. J Cheminform 17, 147. https://doi.org/10.1186/s13321-025-01066-5.

2025 Seal, S. et al. Machine Learning for Toxicity Prediction Using Chemical Structures: Pillars for Success in the Real World. Chem. Res. Toxicol. 38, 759–807 https://doi.org/10.1021/acs.chemrestox.5c00033.

2024 Seal, S. et al. Improved Detection of Drug-Induced Liver Injury by Integrating Predicted In Vivo and In Vitro Data. Chem. Res. Toxicol. 37, 1290–1305. https://doi.org/10.1021/acs.chemrestox.4c00015.

2023 Gupta, N. et al. *CXCR4 Expression Is Elevated in TNBC Patient Derived Samples and Z-Guggulsterone Abrogates Tumor Progression by Targeting CXCL12/CXCR4 Signaling*

Axis in Preclinical Breast Cancer Model. Environmental Research. https://doi.org/10.1016/j.envres.2023.116335.

2023 Martis, E.A.F., et al. *Understanding Protein-Ligand Interactions Using State-of-the-Art Computer Simulation Methods*. In Cheminformatics, QSAR and Machine Learning Applications for Novel Drug Development. Elsevier. https://doi.org/10.1016/B978-0-443-18638-7.00015-3.

Education

2025–2027 **Master in Chemoinformatics for Organic Chemistry**
Universidade Nova de Lisboa, PT & Université de Strasbourg, FR
Current focus: Cheminformatics, Supramolecular Photochemistry, and Multi-omics

2021–2025 **Bachelor of Pharmacy**
Bombay College of Pharmacy, Mumbai, IN
Key courses: Medicinal Chemistry, Pharmacokinetics, Pharmacology, Novel Drug Delivery Systems, Organic Chemistry, Pharmacognosy and Phytochemistry.
Thesis on Foundational Methodologies for Initiating *Caenorhabditis elegans* Research: A Report on Laboratory Setup, Protocol Validation and Preliminary Longevity Assay

Honours

2025 **Erasmus Mundus Joint Master - ChEMoinformatics+**
The European Commission
Full Scholarship for Masters program in Chemoinformatics for Organic Chemistry.

2024 **Gold Medal & Fellowship**
Anveshan, Association of Indian Universities
National Winner in Health Sciences with a $1000 Fellowship.

2023 **Silver Medal & Fellowship**
Avishkar Research Convention, Governor of Maharashtra
Fellowship of $350 for the work on *FragmentBERT: Masked Language Models are Fragment Based Drug Designers*

Volunteering

2023–2024 **Teaching Volunteer**
U&I, Our Lady's Home, Mumbai, IN